

Solution Architecture

Project Design Document - Rising Waters

1. Entity Relationship (ER) Diagram

This diagram represents the entities involved in the Rising Waters tracking and predictions framework:

erDiagram

```

    USER ||--o{ WEATHER_DATA : submits
    WEATHER_DATA ||--|| PREDICTION : generates
    MODEL ||--o{ PREDICTION : predicts
  
```

```

    USER {
        int user_id PK
        string username
        string email
        string role
    }

    WEATHER_DATA {
        int data_id PK
        int user_id FK
        float cloud_cover
        float annual_rainfall
        float jan_feb_rain
        float mar_may_rain
        float jun_sep_rain
    }

    MODEL {
        int model_id PK
        string model_name
        string algorithm_type
        float accuracy
    }

    PREDICTION {
        int prediction_id PK
        int data_id FK
        int model_id FK
        int prediction_result
        float confidence_probability
        string prediction_date
    }
  
```

2. Solution Architecture

The solution uses a standard Model-View-Controller (MVC) serverless design:

1. User Interface (View):

- Renders interactive Bootstrap 5 CSS layouts on the browser.
- Collects variables using standard floating label components.
- Restricts invalid forms using Javascript validations before hitting the server.

2. Web Server Backend (Controller):

- Built with Flask (Python).
- Validates ranges, checks logical rain distribution rules, and flashes validation errors.
- Normalizes feature sets using pre-calculated training metrics.
- Redirects routing paths dynamically based on prediction outputs.

3. Machine Learning Model (Model):

- Standard Supervised Decision Tree Classifier rules compiled directly into a Python function.
- Computes logistic probability curves to provide prediction confidence percentages without loading heavy dependencies.

4. Cloud Infrastructure:

- Hosted on Vercel utilizing Serverless Python Functions, allowing the application to scale dynamically under load.